TECHNICAL EVALUATION — DENX ELECTRIC DEEP FRYER

1. Evaluation output

Field	Value
Report	20 electric deep fryer.pdf (45 pages)
Standards	IEC 60335-1:2010 (Part 1); IEC 60335-2-37:2002 (Part 2 2-37)
Created	2026-11-03 05:44 UTC

2. Report identity

Field	Value
Report number	ED251104331L
Laboratory	EMTEK (DONGGUAN) CO., LTD.
Report date	Nov. 21, 2025
Product declared	Electric deep fryer, DENX
Declared model	20
Standards claimed	US IEC 60335-1:2010; US IEC 60335-2-37:2002
Standards evaluated	US IEC 60335-1:2010; US IEC 60335-2-37:2002

3. Summary

Field	Value
Final verdict	REPORT CORRECTION / REPLACEMENT REQUIRED — PRODUCT COMPLIANCE NOT DETERMINED — CERTIFICATION HOLD
Reason	The report uses requirements for other appliance types, omits essential fryer-specific assessments and lacks referenced test data. Conflicting motor and supply-cord records prevent reliable identification of the tested construction; the recorded 550 °C glow-wire test also does not use the specified 650 °C condition.
Recommended action	Obtain a complete, corrected laboratory report identifying the tested construction and documenting all applicable inspections and tests, including any necessary retesting, before certification proceeds.
Conclusion	The supplied report does not demonstrate compliance of model 20 with US IEC 60335-1:2010 as modified by US IEC 60335-2-37:2002. Product non-conformity cannot be conclusively established from the conflicting component records alone, but the missing and incorrectly applied evidence blocks a certification decision.
Findings	31 total — Critical 4, High 27, Medium 0, Low 0

4. Findings

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
Critical	—	Compliance with ISO 2768-1 can only be confirmed against the text of that standard.	The report states testing to ISO 2768-1, but ISO 2768-1 was not supplied for this review, so its requirements were not evaluated.	Upload ISO 2768-1 and re-run the evaluation, or confirm it is outside the certification scope.	5	Report: NOT FOUND Standard: requirement not in supplied standards
Critical	4; 5.2	Relevant compliance tests are carried out on the specified sample or permitted separate samples.	The report refers to several appended result tables that are not present in the supplied 45-page report; PDF page 44 also repeats printed page number 43. The integrity and completeness of the test record cannot be established.	Obtain the complete, correctly paginated laboratory report and all referenced result tables.	PDF pp.1,11,13–14,41,43–45	Report: verified p.11 Standard: verified in IEC 60335-1:2010 p.26
Critical	19.7; 19.10	Appliances incorporating motors are subjected to applicable motor abnormal-operation tests, including the series-motor test.	The component table lists a 2 000 W series motor, while the report states that there is no such motor and marks motor tests N. The tested construction cannot be reconciled.	Confirm the tested appliance's motor configuration and provide the applicable motor-test evidence or a corrected component list.	PDF pp.16–17,44	Report: verified p.17 Standard: verified in IEC 60335-1:2010 p.55
Critical	25.8; 25.10	Supply-cord conductors must meet the applicable minimum cross-sectional area, and a class I cord must include the specified earthing core.	The report records 0.75 mm², but the component list states a two-core 0.5 mm² cord. At the declared 2 000 W and 220–240 V, the minimum applicable conductor area is 0.75 mm² even under the short portable-cord allowance; the two-core listing also conflicts with the claimed class I earthing arrangement.	Identify the actual cord and cores and demonstrate compliance or replace and retest the cord.	PDF pp.1,5,33,44	Report: NOT FOUND Standard: verified in IEC 60335-1:2010 p.82
High	3.1.9; 5.103; 11.7	The fryer is initially filled with unused vegetable oil, operated at the minimum indicated level, and tested until steady conditions are established.	The heating record refers to durations for other appliance types and does not establish the oil type, oil level, control setting or attainment of steady conditions.	Document the fryer-specific operating conditions and repeat affected tests if those conditions were not used.	PDF pp.11–12,43	Report: verified p.12 Standard: verified in IEC 60335-2-37:2002 p.14
High	7.103	The fryer must be marked with minimum and maximum oil levels.	The report does not assess the required oil-level markings; the marking-plate copy alone does not establish whether they appear elsewhere on the appliance.	Inspect and record both oil-level markings on the tested fryer.	PDF pp.2,5–6	Report: no quote given Standard: verified in IEC 60335-2-37:2002 p.20
High	7.12; 7.12.1	The instructions must contain fryer-specific warnings, maximum batch load, user-maintenance instructions and the water-jet prohibition.	A general instruction pass refers to a product manual, but neither the manual nor an assessment of the specified fryer warnings and batch load is supplied.	Provide and assess the tested model's complete instructions against the fryer-specific requirements.	PDF pp.7–9	Report: verified p.7 Standard: verified in IEC 60335-2-37:2002 p.16
High	10.1	Stabilized power input at rated voltage and normal operation must remain within the applicable Table 1 deviations.	A pass refers to an absent table; measured power input, test voltage and deviation are unavailable.	Provide the referenced power-input measurements and applicable deviations.	PDF pp.1,11	Report: verified p.11 Standard: verified in IEC 60335-1:2010 p.38
High	11.2	Heating and combined appliances normally placed on a floor or table are placed in a test corner unless a particular-part provision changes the installation.	The report records a floor placement but also passes a test away from the test-corner walls; the applicable fryer installation and test-corner arrangement are not established.	Document the actual installation and repeat the heating test using the applicable test-corner arrangement if necessary.	PDF p.11	Report: verified p.11 Standard: verified in IEC 60335-1:2010 p.39
High	11.3; 11.8	Oil temperature is measured at the specified location and must not exceed 200 °C during normal heating.	The temperature table contains no oil-temperature result or measurement position, although the general heating result is passed.	Measure and report maximum oil temperature at the specified location under the required heating conditions.	PDF pp.11–12,43	Report: verified p.12 Standard: verified in IEC 60335-2-37:2002 p.24

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
High	13.2; 13.3; 15.3; 16.2; 16.3	The appliance must meet applicable leakage-current and electric-strength requirements, including after humidity treatment.	The report claims passes but the referenced leakage-current, electric-strength and humidity tables are absent; measured currents, test voltages and humidity conditions cannot be verified.	Supply the complete test tables, conditions and results and reassess them against the applicable limits.	PDF pp.13–14	Report: verified p.14 Standard: verified in IEC 60335-1:2010 p.50
High	14; 29.1	Clearances must meet the applicable minimum values or, where permitted, withstand the impulse-voltage test.	The report claims passes for transient overvoltages and clearances but the referenced clearance table is absent, and no impulse-test details are supplied.	Provide the clearance measurements, applicable limits and any required impulse-test results.	PDF pp.13,41	Report: verified p.41 Standard: verified in IEC 60335-1:2010 p.94
High	15.1.1	IPX0 through IPX4 fryers are subjected to the specified splash test for 5 min.	The report identifies IPX0 but marks the ingress test N and provides no fryer-specific splash-test results.	Perform and document the specified 5 min splash test and subsequent checks.	PDF pp.5,13	Report: verified p.13 Standard: verified in IEC 60335-2-37:2002 p.24
High	15.2	Fryers must be constructed so that normal-use liquid spillage does not affect electrical insulation.	The spillage requirement and overfilling test are marked N without a fryer-specific justification; a separate electric-strength pass does not establish the spillage treatment.	Perform and document the applicable spillage assessment and subsequent insulation checks.	PDF pp.13–14	Report: verified p.13 Standard: verified in IEC 60335-2-37:2002 p.26
High	19.2; 19.3; 19.13	The fryer is subjected to the prescribed abnormal oil conditions, and applicable oil temperatures must not exceed 230 °C.	The report claims abnormal-operation passes using an empty-appliance entry but does not record the two fryer-specific oil conditions or oil temperatures.	Perform and document the fryer-specific abnormal tests, including oil temperatures and ignition observations.	PDF pp.15,19	Report: verified p.15 Standard: verified in IEC 60335-2-37:2002 p.30
High	19.5	Applicable class I appliances with tubular sheathed or embedded heating elements undergo the element-to-sheath fault test, subject to specified exceptions.	The test is marked N without identifying the heating-element construction or establishing an applicable exception.	Identify the element construction and document the test or the applicable exception.	PDF pp.16,24	Report: verified p.16 Standard: verified in IEC 60335-1:2010 p.53
High	19.11.2; 19.12	Applicable electronic fault conditions are evaluated and, where safety depends on a miniature fuse-link, its protection is verified.	The report records passes without identifying the circuits, simulated faults, measured fuse current or fuse rating; it also omits assessment of the listed partial turn-on fault condition.	Provide circuit identification, fault-by-fault results and applicable fuse-protection measurements.	PDF pp.17–19	Report: verified p.19 Standard: verified in IEC 60335-1:2010 p.56
High	22.101	Thermal cut-outs protecting heating circuits must provide the specified non-self-resetting and all-pole disconnection characteristics.	Report clause 22.101 instead addresses fixed water boilers; it does not assess the fryer thermal-cut-out requirement.	Identify the heating-circuit protective devices and assess their construction and operation against 22.101.	PDF pp.22,27	Report: verified p.27 Standard: verified in IEC 60335-2-37:2002 p.32
High	22.102; 22.103	Red indications are restricted to danger or alarm functions, and hot-oil spillage onto parts exceeding 300 °C must be adequately prevented.	The report entries numbered 22.102 and 22.103 address an espresso coffee-maker and a feeding-bottle heater, not the fryer requirements.	Inspect and document the fryer's indication colours and protection against hot-oil contact with hot parts.	PDF p.27	Report: verified p.27 Standard: verified in IEC 60335-2-37:2002 p.34
High	22.108; 22.109	Heating elements must disconnect under the specified removal or swing-up conditions, and the fryer must provide the specified oil-surge allowance.	The report's entries for these clause numbers concern boiling-water appliances and kettles; no fryer element-disconnection assessment or surge-volume-to-batch-load measurement is recorded.	Assess element disconnection and measure the fryer's surge allowance against its recommended batch load.	PDF pp.27–28	Report: verified p.28 Standard: verified in IEC 60335-2-37:2002 p.36
High	22.111	Fryer baskets and applicable movable heating elements must be safely retained in the raised position.	The photographed fryer includes a basket, but the report contains no assessment of its retention in the raised position.	Inspect and manually test the applicable raised-position retention.	PDF p.45	Report: no quote given Standard: verified in IEC 60335-2-37:2002 p.36

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
High	22.114	Portable appliances must prevent small items entering underside openings and reaching live parts, with the specified minimum distances.	The report indicates a mass below 3 kg and describes portable-appliance tests, but contains no underside-opening assessment.	Inspect underside openings and record the applicable distances to live parts.	PDF pp.11,33	Report: no quote given Standard: verified in IEC 60335-2-37:2002 p.36
High	24.1; 24.1.3	Components must meet applicable safety requirements and switches required for compliance must meet the specified cycle requirements.	A 250 V, 3 A switch is listed, but its circuit function, load and suitability in the appliance are not established; the report marks switch compliance N.	Identify the switch function and load and provide component approval or applicable test evidence.	PDF pp.29,44	Report: verified p.44 Standard: verified in IEC 60335-1:2010 p.75
High	25.7	Fryer supply cords must be oil-resistant and at least equivalent to the specified ordinary polychloroprene-sheathed cord.	The report passes an ordinary PVC cord and marks the polychloroprene cord entry N; no evidence of the required oil resistance or equivalence is supplied.	Verify the installed cord against the fryer-specific oil-resistance and cord-type requirements.	PDF pp.32–33,44	Report: verified p.33 Standard: verified in IEC 60335-2-37:2002 p.40
High	25.15; 25.17; 25.18; 25.20	A type Y supply cord requires an adequate anchorage, restricted anchorage access and the specified additional insulation.	The report identifies type Y attachment and a supply cord but marks the anchorage, pull-and-torque test and additional insulation N without justification.	Inspect and test the actual type Y cord arrangement and record the results.	PDF pp.7,31–35	Report: verified p.35 Standard: verified in IEC 60335-1:2010 p.85
High	27.2; 27.3; 27.4	Earthing connections must be secured against loosening and corrosion, and the earth conductor must remain effective if the cord slips.	The report declares a class I cord-connected appliance but marks these earthing assessments N without reasons.	Inspect and document the earthing terminal, corrosion protection and conductor arrangement.	PDF pp.5,31,38–39	Report: verified p.39 Standard: verified in IEC 60335-1:2010 p.90
High	27.5	Earth-continuity resistance must not exceed 0.1 Ω under the specified test conditions.	The report claims a pass but does not record the measured resistance, test current or tested points.	Provide the earth-continuity measurements and test conditions for each relevant accessible metal part.	PDF p.39	Report: verified p.39 Standard: verified in IEC 60335-1:2010 p.91
High	28.1	Applicable screws and nuts used for electrical and earthing connections must withstand the specified mechanical test.	The report marks the torque test N despite claiming electrical and earthing connections; no construction-based exemption is recorded.	Identify applicable fixings and document the required torque tests or valid exemptions.	PDF pp.39–40	Report: verified p.40 Standard: verified in IEC 60335-1:2010 p.91
High	29.2	The fryer microenvironment is pollution degree 3 and insulation CTI is at least 250 unless the specified protection from pollution applies.	The report uses an unrelated conditional pollution-degree entry and provides neither CTI evidence nor justification for a protected microenvironment.	Establish the applicable microenvironment and provide CTI or proof-tracking evidence and creepage measurements.	PDF pp.41,43	Report: verified p.41 Standard: verified in IEC 60335-2-37:2002 p.42
High	30.1	Relevant non-metallic parts must withstand the ball-pressure test at the temperature determined from the applicable temperature rises and minimum temperatures.	The report records an enclosure test at 75 °C and an internal plastic-enclosure rise of 65.0 K, but does not establish which material and location were tested or the basis for the selected temperature.	Identify each tested part and substantiate the ball-pressure temperature from the relevant measured rise.	PDF pp.42–43	Report: verified p.42 Standard: verified in IEC 60335-1:2010 p.102
High	30.2.1	The particular standard specifies a 650 °C glow-wire test under 30.2.1.	The report records the enclosure glow-wire test at 550 °C and passes it; this is below the specified fryer test temperature.	Test the relevant parts at 650 °C or provide evidence supporting an applicable permitted alternative.	PDF p.42	Report: verified p.42 Standard: verified in IEC 60335-2-37:2002 p.42